

Technical Document #480: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.
- Query decomposition breaks complex queries into simpler sub-queries.